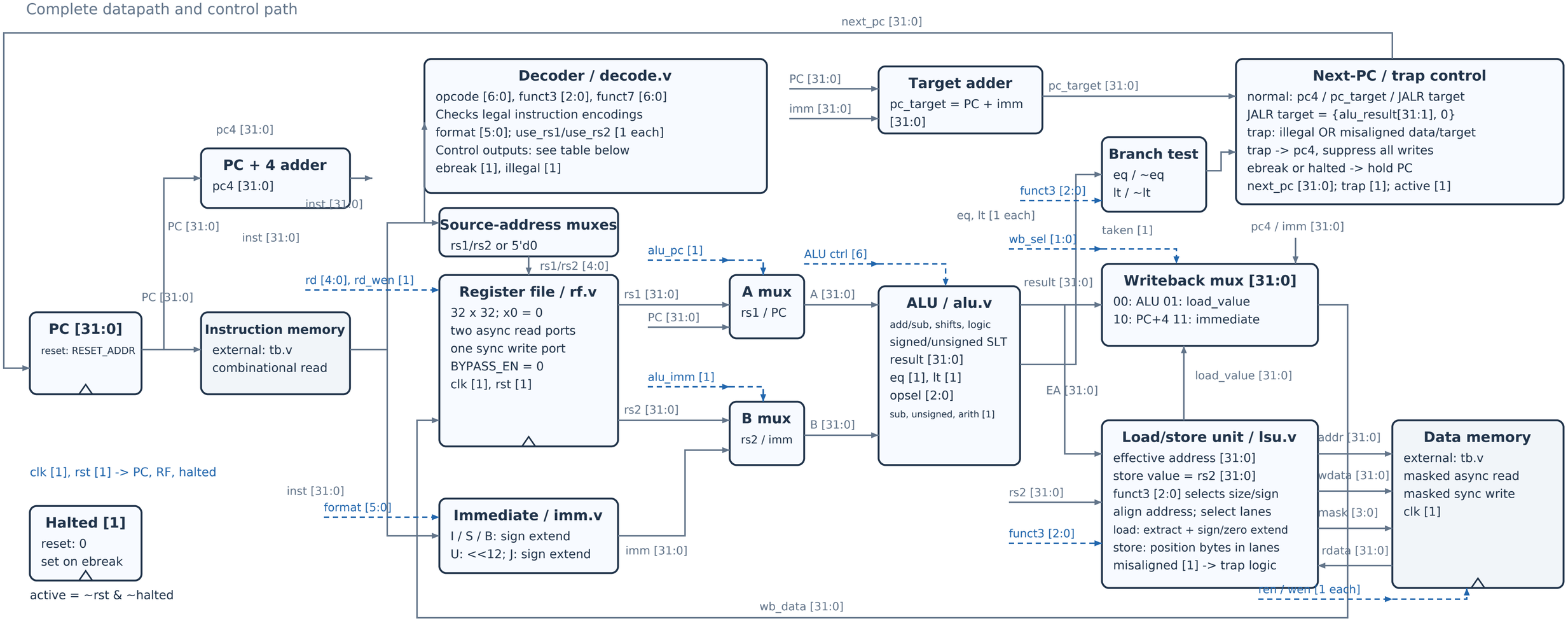
# Single-cycle RV32I processor

Complete datapath and control path






Triangles mark clocked state. Gray memory blocks are external to hart. Unmarked blocks are combinational.

Blue dashed wires are control. Matching named nets are electrically connected; bus widths are shown in brackets.

Retire outputs observe this cycle's instruction, source operands, writeback, PC and next PC; omitted from schematic per assignment.

## Decoder outputs and selection logic

| Instruction | format | ALU A / B | ALU op | wb_sel | RegW | Memory / PC |
|---|---|---|---|---|---|---|
| R arithmetic | 000001 | rs1 / rs2 | funct3 | 00 | 1 | none / pc4 |
| I arithmetic | 000010 | rs1 / imm | funct3 | 00 | 1 | none / pc4 |
| Load | 000010 | rs1 / imm | ADD | 01 | 1 | read / pc4 |
| Store | 000100 | rs1 / imm | ADD | -- | 0 | write / pc4 |
| Branch | 001000 | rs1 / rs2 | compare | -- | 0 | none / target* |
| LUI | 010000 | unused | unused | 11 | 1 | none / pc4 |
| AUIPC | 010000 | PC / imm | ADD | 00 | 1 | none / pc4 |
| JAL | 100000 | unused | unused | 10 | 1 | none / target |
| JALR | 000010 | rs1 / imm | ADD | 10 | 1 | none / ALU&~1 |
| EBREAK / FENCE | 000001 | unused | unused | -- | 0 | hold / pc4 |

* branch target selected only if the comparison is true; otherwise pc4.
sub: R-type SUB only; arith: SRA/SRAI; unsigned: SLTU/SLTIU or BLTU/BGEU.
use_rs1/rs2 [1 each] select inst[19:15]/inst[24:20], or 5'd0, into RF.
rd_wen = RegW & active & ~trap & (rd != 0); rd = inst[11:7].
ren/wen = load/store & active & ~trap; illegal encodings have no side effects.

## Memory lanes, branch conditions, and state control

EA [31:0] = rs1 + immediate; offset [1:0] = EA[1:0].
o_dmem_addr [31:0] = {EA[31:2], 2'b00}.
Byte mask [3:0]: offset 0/1/2/3 -> 0001/0010/0100/1000.
Half mask [3:0]: offset 0/2 -> 0011/1100; word mask = 1111.
Store mux [31:0]: rs2 << 0/8/16/24 selected by offset [1:0].
Load mux [31:0]: rdata >> 0/8/16/24 selected by offset [1:0].
LB/LH sign-extend low 8/16 bits; LBU/LHU zero-extend; LW passes 32 bits.
All lane shifts use constant wiring and muxes, never variable shift operators.
misaligned [1] = half & EA[0] OR word & (EA[1:0] != 0).
BEQ/BNE select eq/~eq; BLT/BLTU select lt; BGE/BGEU select ~lt.
Jump or taken branch traps if target[1:0] != 0 (after JALR clears bit 0).
Trap policy: suppress register/memory writes; continue at pc4 (no CSRs).
EBREAK: assert halt this cycle; hold PC; latch halted on rising clock.
Halted: no memory access, register writes or retirement until reset.
Synchronous reset: PC <- RESET_ADDR; RF <- 0; halted <- 0.

## Named control nets: source -> destination (port connections)

decode.use_rs1/use_rs2 [1 each] -> source-address mux selects.
decode.format [5:0] -> imm.i_format; inst [31:0] -> decode.i_inst / imm.i_inst.
decode.alu_pc / alu_imm [1 each] -> A / B mux selects.
decode.alu_op [2:0], sub/unsigned/arith [1 each] -> ALU ctrl [5:0].
decode.wb_sel [1:0] -> writeback mux; decode.reg_write [1] -> rd_wen gate.
decode.load/store [1 each] -> ren/wen gates and data-alignment trap gate.
decode.branch/jal/jalr [1 each] -> redirect / target mux in next-PC logic.
decode.ebreak [1] -> halted set input and next-PC hold select.
decode.illegal [1] -> trap OR; trap [1] -> write/access gates and next-PC mux.

## Named data nets and next-PC mux inputs

PC [31:0] -> instruction address, PC+4 adder, target adder, ALU A mux.
imm [31:0] -> ALU B mux, target adder, writeback input 11.
pc4 [31:0] -> next-PC default and writeback input 10.
alu_result [31:0] -> JALR target clear-bit-0 wiring, LSU, writeback input 00.
rs2_data [31:0] -> LSU store-data input and ALU B mux.
LSU.misaligned [1] / target[1:0] / PC[1:0] -> trap detection.
redirect [1] = jal | jalr | (branch & branch_condition).
next_pc = (halted | ebreak) ? PC : trap ? pc4 : redirect ? target : pc4.
halted_next [1] = halted | ebreak; clk/rst [1 each] -> all state elements.

Source: schematics/*.svg (editable in Inkscape). Implementation: rtl/{hart,decode,alu,rf,imm,lsu}.v